MTH 622

Quiz No.2

Solution by Sufyan

JUTT

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MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 07:37 PM, 06 December 2023

Question # 1 of 10 (Start time: 07:37:09 PM, 06 December 2023) Total Marks: 1

When a particle moves along a straight path, then the particle has

Select the correct option

- none of these
- both tangential and centripetal acceleration
- centripetal acceleration only
- tangential acceleration only

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 07:37 PM, 06 December 2023

Question # 2 of 10 (Start time: 07:37:47 PM, 06 December 2023) **Total Marks:** 1

The radial and transverse components of velocity are

Select the correct option

Reload Math Equations

- (●) $\dot{r},\ r\dot{\theta}$
- ( ) $\dot{r},\ \theta\dot{\theta}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 07:37 PM, 06 December 2023

Question # 3 of 10 (Start time: 07:38:41 PM, 06 December 2023) **Total Marks:** 1

Which of the following is the simplest type of motion along a straight line path.

Select the correct option

- ( ) curvilinear Motion
- (●) rectilinear motion
- ( ) plane motion
- ( ) none of these

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 07:37 PM, 06 December 2023

Question # 4 of 10 (Start time: 07:39:09 PM, 06 December 2023) **Total Marks:** 1

Evaluate $\int_c (x^2 dx + y^2 dy)$ along the line segment joining $(1, 0)$ to $(1, 1)$.

Select the correct option

Reload Math Equations

- ○ $\frac{1}{2}$
- ◉ $\frac{1}{3}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 07:37 PM, 06 December 2023

Question # 5 of 10 (Start time: 07:40:46 PM, 06 December 2023) **Total Marks:** 1

If a force acts on a particle of constant mass m and $\vec{v}_1$, $\vec{v}_2$ are velocities at times t_1 and t_2, respectively. Then the work done by the force is obtained as

Select the correct option

Reload Math Equations

- ◉ The change in kinetic energies, i.e., $\frac{1}{2}m\vec{v}_2^2 - \frac{1}{2}m\vec{v}_1^2$
- ○ The sum of kinetic energies, i.e., $\frac{1}{2}m\vec{v}_2^2 + \frac{1}{2}m\vec{v}_1^2$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 07:37 PM, 06 December 2023

Question # 6 of 10 (Start time: 07:41:26 PM, 06 December 2023) Total Marks: 1

If k is curvature of a space curve then radius of curvature is

Select the correct option

Reload Math Equations

- $\sqrt{k}$
- $\frac{1}{k}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 07:37 PM, 06 December 2023

Question # 7 of 10 (Start time: 07:42:31 PM, 06 December 2023) Total Marks: 1

Newton's first law is also called

Select the correct option

- law of inertia
- law of conservation of momentum

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 07:37 PM, 06 December 2023

Question # 8 of 10 (Start time: 07:43:04 PM, 06 December 2023) **Total Marks:** 1

The radial component of velocity for a particle moving in circular path is

Select the correct option

- radius itself
- variable
- zero
- constant

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 07:37 PM, 06 December 2023

Question # 10 of 10 (Start time: 07:45:08 PM, 06 December 2023) **Total Marks:** 1

Evaluate $\int_c (x+y)dy$ along the line segment joining $(0,0)$ to $(0,1)$

Select the correct option

Reload Math Equations

- $\frac{1}{3}$
- $\frac{1}{2}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:02 PM, 06 December 2023

Question # 1 of 10 (Start time: 08:02:43 PM, 06 December 2023) **Total Marks:** 1

If k is curvature of a space curve then radius of curvature is

Select the correct option

Reload Math Equations

- $\sqrt{k}$
- $\frac{1}{k}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:02 PM, 06 December 2023

Question # 2 of 10 (Start time: 08:03:18 PM, 06 December 2023) **Total Marks:** 1

If a particle of mass 2 units has position vector $\vec{r} = t^3\hat{i} + 8t\hat{j} - 9t\hat{k}$ then find the force acting on the particle.

Select the correct option

Reload Math Equations

- $12t$
- $12t\hat{i}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:02 PM, 06 December 2023

Question # 3 of 10 (Start time: 08:04:39 PM, 06 December 2023) Total Marks: 1

If a force acts on a particle of constant mass m and $\vec{v}_1, \vec{v}_2$ are velocities at times t_1 and t_2, respectively. Then the work done by the force is obtained as

Select the correct option

Reload Math Equations

- (●) The change in kinetic energies, i.e., $\frac{1}{2}m\vec{v}_2^2 - \frac{1}{2}m\vec{v}_1^2$
- ( ) The sum of kinetic energies, i.e., $\frac{1}{2}m\vec{v}_2^2 + \frac{1}{2}m\vec{v}_1^2$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:02 PM, 06 December 2023

Question # 4 of 10 (Start time: 08:05:10 PM, 06 December 2023) Total Marks: 1

Evaluate $\int_c (x^2dx + y^2dy)$ along the line segment joining $(1,0)$ to $(1,1)$.

Select the correct option

Reload Math Equations

- (●) $\frac{1}{3}$
- ( ) $\frac{1}{2}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:02 PM, 06 December 2023

Question # 5 of 10 (Start time: 08:07:05 PM, 06 December 2023) **Total Marks:** 1

Which of the following objects has more inertia?

Select the correct option

Reload Math Equations

- (●) an object with mass $40kg$
- ( ) an object with mass $30kg$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:02 PM, 06 December 2023

Question # 6 of 10 (Start time: 08:07:45 PM, 06 December 2023) **Total Marks:** 1

Which of the following is the simplest type of motion along a straight line path.

Select the correct option

- (●) rectilinear motion
- ( ) curvilinear Motion
- ( ) none of these
- ( ) plane motion

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:02 PM, 06 December 2023

Question # 7 of 10 (Start time: 08:08:45 PM, 06 December 2023) **Total Marks:** 1

According to Newton's second law of motion the net force applied on a body is equal to the rate of change of momentum of body.

Select the correct option

- ○ false
- ◉ true

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:02 PM, 06 December 2023

Question # 8 of 10 (Start time: 08:09:31 PM, 06 December 2023) **Total Marks:** 1

When a particle moves along a straight path, then the particle has

Select the correct option

- ○ centripetal acceleration only
- ○ both tangential and centripetal acceleration
- ○ none of these
- ◉ tangential acceleration only

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:02 PM, 06 December 2023

Question # 9 of 10 (Start time: 08:10:03 PM, 06 December 2023) **Total Marks:** 1

If $\vec{T}$ is a unit vector to a space curve C and S is the arc length then $\frac{d\vec{T}}{dS}$ is

Select the correct option

Reload Math Equations

- (●) perpendicular to $\vec{T}$
- ( ) parallel to $\vec{T}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:02 PM, 06 December 2023

Question # 10 of 10 (Start time: 08:10:42 PM, 06 December 2023) **Total Marks:** 1

According to first law of motion if some external force is acting on a body then the velocity of object is constant.

Select the correct option

- (●) false
- ( ) true

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:11 PM, 06 December 2023

Question # 1 of 10 (Start time: 08:11:57 PM, 06 December 2023) **Total Marks:** 1

Evaluate $\int_c (x+y)dy$ along the line segment joining $(0,0)$ to $(0,1)$

Select the correct option

Reload Math Equations

- ○ $\frac{1}{3}$
- ◉ $\frac{1}{2}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:11 PM, 06 December 2023

Question # 2 of 10 (Start time: 08:12:38 PM, 06 December 2023) **Total Marks:** 1

The product of force and displacement is called

Select the correct option

- ○ energy
- ◉ work done

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:11 PM, 06 December 2023

Question # 3 of 10 (Start time: 08:13:21 PM, 06 December 2023) Total Marks: 1

Newton's first law is also called

Select the correct option

- [ ] law of conservation of momentum
- [x] law of inertia

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:11 PM, 06 December 2023

Question # 4 of 10 (Start time: 08:13:41 PM, 06 December 2023) Total Marks: 1

The normal component of acceleration is zero when motion of particle is

Select the correct option

- [ ] translation
- [ ] rotational
- [ ] curvilinear
- [x] rectilinear

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:11 PM, 06 December 2023

Question # 5 of 10 (Start time: 08:14:26 PM, 06 December 2023) **Total Marks:** 1

The radial component of velocity for a particle moving in circular path is

Select the correct option

- constant
- radius itself
- variable
- zero

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:11 PM, 06 December 2023

Question # 6 of 10 (Start time: 08:14:56 PM, 06 December 2023) **Total Marks:** 1

A particle moves in a plane in such a way that at any time t its distance from the origin is $r = 2t + 3t^2$ and its angle with positive x-axis is $\theta = \sqrt{t}$. Find the radial and transverse components of velocity.

Select the correct option

Reload Math Equations

- $2 + 6t$ and $(\sqrt{t} + \frac{3}{2}t^{\frac{3}{2}})$
- $2 + 6t$ and $\frac{1}{2\sqrt{t}}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:11 PM, 06 December 2023

Question # 7 of 10 (Start time: 08:16:12 PM, 06 December 2023) Total Marks: 1

The radial and transverse components of velocity are

Select the correct option

Reload Math Equations

- (●) $\dot{r},\ r\dot{\theta}$
- ( ) $\dot{r},\ \theta\dot{\theta}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:11 PM, 06 December 2023

Question # 8 of 10 (Start time: 08:16:28 PM, 06 December 2023) Total Marks: 1

Which of the following is the simplest type of motion along a straight line path.

Select the correct option

- (●) rectilinear motion
- ( ) plane motion
- ( ) none of these
- ( ) curvilinear Motion

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:11 PM, 06 December 2023

Question # 9 of 10 (Start time: 08:17:01 PM, 06 December 2023) **Total Marks:** 1

If k is curvature of a space curve then radius of curvature is

Select the correct option

Reload Math Equations

- ○ $\sqrt{k}$
- ● $\frac{1}{k}$

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (Quiz No 2) Quiz Start Time: 08:11 PM, 06 December 2023

Question # 10 of 10 (Start time: 08:17:36 PM, 06 December 2023) **Total Marks:** 1

According to Newton's second law of motion the net force applied on a body is equal to the rate of change of momentum of body.

Select the correct option

- ○ false
- ● true

Click to Save Answer & Move to Next Question

A person is pushed forward into his seat belt when car stops. Which of Newton's laws explains this motion?

lect the correct option

○	Newton's second law of motion
○	Newton's third law of motion
○	None of these
○	Newton's first law of motion

Click to Save Answer & Move to Next Questi

Question # 8 of 10 (Start time: 05:01:26 PM, 06 December 2023)

Evaluate $\int_c (\cos^2 x \sin y)dy$ along the line segment joining $(0,0)$ to $(0,\pi)$.

Select the correct option

○	2
○	-2

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